#1 For each variable, there's one H_0 and H_a

H_0: Null-hypothesis: there is no relationship between a variable and the outcome (sales)

H_a: there is some relationship between the variable and the outcome (sales)

In multiple linear regression, each individual p-value tests a separate null hypothesis that assumes all *other* variables are already in the model.

Based on the fact that a few p-values are < 0.05, the variables indicate at least some relationship with the outcome. Thus, the null-hypothesis can be rejected for those.

However, the p-value for the newspaper is high (0.8599) and could be a subject for further exploration if it correlates with other variables, is insignificant and could be removed, or has an interaction effect with TV or Radio.

#2.

KNN - classifier

- predicts a label
- majority KNN becomes the outcome

KNN - regression

- predicts a number
- performs ok at low p's
- performs worse than the linear model with the right fit
- averages the KNN values

- curse of dimensionality

#3. a)

High School : $50 + 20 \cdot GPA + 0.07 \cdot IQ + 0.01(GPA \cdot IQ)$

College : $50 + 20 \cdot GPA + 0.07 \cdot IQ + 0.01(GPA \cdot IQ)$
$+ 35 - 10 \cdot GPA$

$0 \leq GPA \leq 4$

College graduates earn more if $GPA < 3.5$
earn less if $GPA > 3.5$

iii → High School graduates earn more if $GPA > 3.5$
earn less if $GPA < 3.5$

Statement iii is correct

b) $IQ = 110$ $GPA = 4$ $College = 1$

$salary = 50 + 20 \cdot GPA + 0.07 \cdot IQ + 35 \cdot College + 0.01(GPA \cdot IQ) - 10(GPA \cdot College)$

$50 + 20 \cdot 4 + 0.07 \cdot 110 + 35 + 0.01 \cdot 4 \cdot 110 - 10 \cdot 4$

salary = \$137.1k

c) Insufficient to conclude that there is very little evidence of an interaction effect between GPA & IQ solely based on the small coefficient of 0.01.

The IQ itself contributes very little to the model, so the interaction term still can be significant enough to be included and should be evaluated further with other statistical means such as colinearity or p-values.

Also, typically $\sim 70 < IQ < 150$, $0 < GPA < 4$, so even small IQ coefficient might add as much as high GPA coefficient given the ranges.

#4. a) Expect training RSS for Linear regression to be higher than cubic regression because cubic allows higher variance and thus its residuals will be lower.

b) The testing RSS for linear reg. will be lower because the new data will follow linear trend better than the overfitted cubic model.

c) Cubic will still fit better for the training RSS due to its flexibility

d) However, there's no enough information about the real curvature to tell which testing RSS would perform better.

#5. Given:

$$\hat{y}_k = x_k \cdot \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} = \frac{\sum_{i'=1}^{n} x_k x_{i'} y_{i'}}{\sum_{i=1}^{n} x_i^2} =$$

treat $\frac{1}{\sum_{i=1}^{n} x_i^2}$ as a single number p

$$= p \sum_{i'=1}^{n} x_k x_{i'} y_{i'} = \sum_{i'=1}^{n} p x_k x_{i'} y_{i'} =$$

$$= \sum_{i'=1}^{n} \frac{x_k x_{i'}}{\sum_{j=1}^{n} x_j^2} \cdot y_{i'}$$

if $\hat{y}_i = \sum_{i'=1}^{n} a_{i'} y_{i'}$, then $a_{i'} = \frac{x_i x_{i'}}{\sum_{j=1}^{n} x_j^2}$

#6. Simple LR formula:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i \qquad \bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

Prove that fit line passes through $(\bar{x}, \bar{y})$

$$\hat{y} = \hat{\beta}_1 x + \bar{y} - \hat{\beta}_1 \bar{x} = \hat{\beta}_1 (x - \bar{x}) + \bar{y}$$

$$= (x - \bar{x}) \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2} + \bar{y}$$

if $x = \bar{x}$, $(x - \bar{x}) = (\bar{x} - \bar{x}) = 0$

then $\hat{y} = 0 + \bar{y} = \bar{y}$

thus, the line passes through $(\bar{x}, \bar{y})$

#7. Assuming $\bar{x} = \bar{y} = 0$

Prove $R^2 = Cor(X, Y)^2$ for Simple Linear Regression

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS} \qquad RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

$$R^2 = \frac{\sum_{i=1}^{n} (y_i - \bar{y})^2 - \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$$Cor(X, Y) = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}} \qquad Cor(X, Y)^2 = \frac{\left(\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2 \sum_{i=1}^{n} (y_i - \bar{y})^2}$$

$$\left(\sum_{1}^{4} x\right)^2 \neq \sum_{1}^{4} x^2$$

Substituting 0's:

$$Cor(X,Y) = \frac{\left(\sum_{i=1}^{n} x_i y_i\right)^2}{\left(\sum_{i=1}^{n} x_i^2\right)\cdot\left(\sum_{i=1}^{n} y_i^2\right)}$$

$$\hat{y}_i = (x_i - \bar{x})\frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} + \bar{y}$$

$$= x_i \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2} = x_i \hat{\beta}_1$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{n}(x_i-\bar{x})^2} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}$$

$$\hat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} x_i y_i$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n}(y_i-\bar{y})^2} = 1 - \frac{\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}{\sum_{i=1}^{n} y_i^2} =$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n} y_i^2 - \sum_{i=1}^{n} 2\cdot y_i \cdot \hat{y}_i + \sum_{i=1}^{n}\hat{y}_i^2}{\sum_{i=1}^{n} y_i^2}$$

$$= \frac{2\sum_{i=1}^{n} y_i \hat{y}_i - \sum_{i=1}^{n}\hat{y}_i^2}{\sum_{i=1}^{n} y_i^2} = \frac{2\hat{\beta}_1 \sum_{i=1}^{n} y_i x_i - \hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n} y_i^2}$$

$$= \frac{1}{\sum_{i=1}^{n} y_i^2}\left(2\hat{\beta}_1 \sum_{i=1}^{n} y_i x_i - \hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2\right) =$$

$$= \frac{1}{\sum_{i=1}^{n} y_i^2}\left(2\hat{\beta}_1 \cdot \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 - \hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2\right) =$$

$$= \frac{1}{\sum_{i=1}^{n} y_i^2}\left(2\hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2 - \hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2\right) = \frac{\hat{\beta}_1^2 \sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n} y_i^2}$$

$$= \left(\frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}\right)^2 \cdot \frac{\sum_{i=1}^{n} x_i^2}{\sum_{i=1}^{n} y_i^2} = \frac{\left(\sum_{i=1}^{n} x_i y_i\right)^2}{\left(\sum_{i=1}^{n} x_i^2\right)^2} \cdot \frac{\cancel{\sum_{i=1}^{n} x_i^2}}{\sum_{i=1}^{n} y_i^2}$$

$$= \frac{\left(\sum_{i=1}^{n} x_i y_i\right)^2}{\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right)}$$

$$Cor(X,Y) = \frac{\left(\sum_{i=1}^{n} x_i y_i\right)^2}{\left(\sum_{i=1}^{n} x_i^2\right)\cdot\left(\sum_{i=1}^{n} y_i^2\right)} = R^2 = \frac{\left(\sum_{i=1}^{n} x_i y_i\right)^2}{\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right)}$$

$$Cor(X,Y) = R^2$$

QED □